Boiler Startup Procedure - SOP-202
Equipment: Boiler B-101 | Document: SOP-202 | Rev: 3.0

1. PURPOSE
This Standard Operating Procedure describes the safe and efficient startup procedure for Boiler B-101 at the Main Processing Unit.
2. PREREQUISITES
- Verify fuel gas pressure between 3.5-4.5 barg at header FG-001
- Ensure feed water tank level is above 65% (FT-101 > 65%)
- Confirm instrument air pressure is at least 6.0 barg
- All safety interlocks have been tested and are operational
- Permit-to-Work (PTW) system clearance obtained
3. STARTUP SEQUENCE
Step 1: Open fuel gas isolation valve V-101 slowly. Monitor pressure gauge PG-101 for stable reading at 3.8 barg.
Step 2: Initiate purge cycle. Run forced draft fan F-101 for 5 minutes at 30% speed to clear any residual gas from the combustion chamber.
Step 3: Set burner management system BMS-101 to AUTO mode.
Step 4: Verify flame scanner FS-101 shows NO FLAME before ignition.
Step 5: Initiate pilot burner ignition. Confirm pilot flame within 3 seconds.
Step 6: Open main fuel valve V-102 gradually. Target drum pressure ramp:
0-10 barg: 0.5 bar/min
10-30 barg: 1.0 bar/min
30-50 barg: 0.75 bar/min
Step 7: When drum pressure reaches 50 barg, switch from Bypass to Main steam line. Align valve V-201 to match header pressure.
4. CRITICAL PARAMETERS
- Drum Level must remain between 40-70% throughout startup
- Superheater outlet temperature should not exceed 450 C
- Furnace pressure must be maintained at -2 to +2 mbar
- NOx emissions must stay below 50 ppm
5. EMERGENCY CONDITIONS
Abort startup immediately if any of the following occur:
- Flame failure detected for more than 4 seconds
- Drum level drops below 25%
- Fuel gas pressure exceeds 5.5 barg or drops below 2.5 barg
- Any safety interlock trips during startup
6. DOCUMENTATION
Record all startup parameters in Log Sheet LS-B101. Notify Shift Supervisor upon reaching stable operating conditions (steady state for 30 minutes).